Sound Becomes a Signal

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Sound Becomes a Signal

> Prepare Sound Becomes a Signal: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Interpret an ordered sequence of sound-**signal sample**s with a stated time interval

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Pictures Become Numbers

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

Look for: Encode and reconstruct a small image using an explicit grid convention

Reference example, not the only acceptable wording: The upper row has black, white, black; the lower row has white, black, white. Reading back gives the supplied two rows.

There are three black pixels among six total positions.

Sensors Are Messengers

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

Look for: Describe a measurement using its quantity, value, unit and conditions

Reference example, not the only acceptable wording: Keep the observed value 24 and approximate time lunchtime; mark the unit, exact time if needed, location, exposure and device identity as unknown or to be checked.

A suitable completed record needs air temperature, value, unit and relevant conditions. Invented degrees C or shade is not an **observation**.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A whisper recorded close to a microphone has larger **sample amplitude**s than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

Concept to explain

Computers receive changing measurements, not human-like hearing.

Term	Meaning
waveform	graph of a signal changing over time
sample	one measurement at one moment
amplitude	size or strength of a signal value
noise	unwanted signal mixed with the target

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Keep the distinction between physical pressure variation, microphone **signal** and stored numerical **sample**s.

The examples use arbitrary nonnegative **signal** values to avoid introducing signed **waveform**s before they are needed; they do not model real speech faithfully. The lesson establishes ordering, time metadata and information loss.

It does not teach a sampling theorem or make hearing-health claims.

Readiness answer

The total does not preserve positions. Different arrangements can contain the same number of black pixels.

A time strip marked 0 through 8 milliseconds and paper **sample**-value cards

A cover sheet to hide omitted time positions

Printed fictional speech requests; recording students or opening accounts is unnecessary

End goal for the pictorial case

By the end, I can connect ordered sound **sample**s to timing and explain what a maximum loses.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Sound Becomes a Signal using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT changing air pressure

2. Process — the deciding step

PROCESS sample the **signal** and compare a clap pattern

3. Output — the result

OUTPUT clap detected, not detected, or uncertain

Key idea

CORE IDEA Computers receive changing measurements, not human-like hearing.

How the mechanism works

UNDER THE HOOD Audio is represented by **sample**s taken over time. **Sample** rate describes how often measurements are taken; **amplitude** describes **signal** strength.

Noise can overlap the target pattern, so detection requires a decision rule or model and an uncertainty policy.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A microphone converts air-pressure changes into a sequence of measured values that can be plotted as a **waveform**.

Step 1

STEP 1 — OBSERVE changing air pressure

Step 2

STEP 2 — TRACE **sample** the **signal** and compare a clap pattern

Step 3

STEP 3 — CONCLUDE clap detected, not detected, or uncertain

Model answer

MODEL REASONING Look for two strong peaks separated by the expected gap. **Noise** may add extra peaks or hide a clap. A safe detector needs thresholds, timing rules, and a not-sure result when **evidence** is weak.

A claim to avoid

DO NOT OVERCLAIM A microphone measures a **signal**; it does not understand a speaker's meaning by itself.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

A microphone records quiet and loud claps at the same distance and settings.

What **waveform** difference is expected? Does it reveal a spoken word?

Exact explanation to model

Why the result follows

The louder clap generally produces larger **amplitude**. **Amplitude** is not word meaning. Distance, gain and **noise** must be controlled.

Explain the added worked example

Explain the added worked example · 1

Sound involves changing pressure in a medium such as air. A microphone responds to that variation and produces a changing **signal**. A digital recording stores measurements of the **signal** at successive times.

The values are ordered: swapping their times changes the represented pattern.

Explain the added worked example · 2

We will use tiny invented sequences to study this idea on paper. Their values are arbitrary **signal** units, not decibels or a test of safe listening levels.

Real audio uses many more **sample**s, so our five-point examples should not be played as if they were a realistic recording of speech.

A recording must include enough information to interpret the values, including their time spacing and encoding. The sequence 0,2,0,2,0 does not by itself say how quickly the changes happened.

This is similar to needing image dimensions as well as pixel values.

Explain the added worked example · 3

There are five **sample**s with four gaps between the first and last. Each gap is 1 millisecond, so the elapsed time from first to last is 4 milliseconds.

If the same five values were spaced 2 milliseconds apart, they would span 8 milliseconds. Equal value lists with different timing do not describe the same time pattern.

Key terms

Term	Meaning
Signal	A changing physical or recorded quantity that can carry information
Sample	A recorded value of a signal at a particular time in this lesson
Sampling interval	The time between successive recorded samples
Waveform	A representation of how a signal value changes over time
Transcript	A text representation of spoken words, which may contain errors or omit sound details

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

The largest value in 0,2,0,2,0 is 2. The largest value in 0,0,2,0,0 is also 2. The first sequence reaches 2 twice at different times, while the second does so once.

Keeping only the maximum loses that distinction, just as counting black pixels lost their arrangement.

Model the reasoning aloud · 4

Sparse measurements can also miss changes. If we inspect the first sequence only at times 0, 2 and 4 milliseconds, all retained values are 0. The two intervening values of 2 disappear from the record.

It would be incorrect to conclude from these retained **sample**s alone that the **signal** stayed at zero between them.

Model the reasoning aloud · 5

More frequent sampling can preserve faster changes under suitable recording conditions, but sampling frequency alone does not make every recording clear.

Microphone placement, background sound, clipping and the recording format can also matter. We are studying one source of missing information, not a complete audio engineering specification.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sound Becomes a Signal

A fictional microphone display records five **sample**s during a short classroom tap.

The facts we will use

Supplied facts

Times are 0,1,2,3,4 ms in order.

Supplied facts

Signal A: 0,3,0,1,0. **Signal** B: 0,1,0,3,0. Toy values are signed measurement units; these happen to be nonnegative.

AI ACADEMY / AI-1/6

Same peak, different sample timing

	0 ms	1 ms	2 ms	3 ms	4 ms
Signal A	0	3	0	1	0
Signal B	0	1	0	3	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/6

Sound Becomes a Signal

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Timed sampling

Associate every measurement with its sampling time.

↓

2. Signal A

Its largest sample 3 occurs at 1 ms.

↓

3. Signal B

Its largest sample 3 occurs at 3 ms.

↓

4. Maximum summary

Both maxima are 3, but the strongest recorded tap arrives at different times.

CHANGE ONE THING

Retain only samples at 0,2,4 ms.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Timed sampling	Associate every measurement with its sampling time.
2. Signal A	Its largest sample 3 occurs at 1 ms.
3. Signal B	Its largest sample 3 occurs at 3 ms.
4. Maximum summary	Both maxima are 3, but the strongest recorded tap arrives at different times.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The sequences contain the same values in different orders. Keeping the maximum preserves peak recorded value while losing its timing.

A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

What this example does not establish

Keep the stated limit

These sparse toy values cannot establish perceived loudness, pitch or what happened between sampling times.

Change one thing in the pictured case

Change one thing

Change: Retain only **sample**s at 0,2,4 ms.

Predicted result

Prediction: Both retained sequences are 0,0,0.

Why the result follows

Reason: The selected times omit both nonzero **sample**s from each **signal**.

Ask while showing the picture

Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Sound Becomes a Signal, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A microphone records quiet and loud claps at the same distance and settings.

2. State the decision to explain

Write the question to explain: What **waveform** difference is expected? Does it reveal a spoken word?

3. Reveal the worked reasoning

Reveal after **prediction**s: The louder clap generally produces larger **amplitude**. **Amplitude** is not word meaning. Distance, gain and **noise** must be controlled.

Hold back the independent task

Keep for an independent attempt: Two recordings contain the same word: one quiet and one loud. Which device receives the sound? Does a larger wave automatically mean a different word? Explain.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

The interval is 2 ms and first-to-last duration 8 ms. Keeping 0,4,8 ms retains 0,0,0 and misses the 3 and 1 at 2 and 6 ms. A **transcript** can preserve estimated words but may omit pauses or background sounds.

Correct arithmetic on a transcribed number does not establish that the number matches the recording.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve equal pixel counts with different arrangements. Replace position in space with position in time.
7 to 20 minutes	Trace the recording/transcript/processing diagram. Explain the invented units and times.

Stage	Teacher action
20 to 33 minutes	Count gaps physically between five sample cards. Compare the same values at two time intervals.
33 to 45 minutes	Complete W1 and W2, covering omitted positions and naming lost changes.
Session 2 start 0 to 8 minutes	Compare the equal-maximum sequences. Require a specific difference, not only less information.
8 to 23 minutes	Complete W3 and practise constructing two records that agree at retained times.
23 to 38 minutes	Collect W4 independently, checking all seven positions and the three retained constraints.
38 to 45 minutes	Discuss W5 and the exit question. Close the unit by connecting evidence, representation, measurement, fitting, pixels and samples.

Reduce help deliberately

Completion practice

Completion task: Retain only times 0,4,8 from W1. Write the retained values. Give one claim about the original **signal** that these retained values do not justify.

Reasoning to listen for

Teacher check: The retained values are 0,0,0. They do not justify saying the **signal** was always zero or had no changes between those times; the original had values 3 and 1 at omitted times.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Sound as **sample**d information. Proposed explanation: “Five **sample**s make five gaps”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: Place five cards in a row and point to each space between them. The span ends at the last **sample**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Does the all-zero retained record prove the full **signal** was silent? Use the picture.

Reasoned teacher answer

No. The full supplied **signal**s contain nonzero values at 1 and 3 ms; those measurements were discarded.

Changed-case reasoning: Both retained sequences are 0,0,0. The selected times omit both nonzero **sample**s from each **signal**.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Sound Becomes a Signal investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Build a clap-pattern detector; test it with background **noise**.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR Louder always means a different word.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Five **sample**s make five gaps: Place five cards in a row and point to each space between them. The span ends at the last **sample**.

Equal recorded zeros prove the **signal** stayed zero: Reveal hidden **sample** cards between retained zeros. Construct both changing and unchanging originals.

A **transcript** is the recording in another file type: Compare the information each representation retains. Text estimates words and need not preserve sound details or exact wording.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Sound Becomes a Signal questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

The louder clap generally produces larger **amplitude**. **Amplitude** is not word meaning. Distance, gain and **noise** must be controlled.

Independent question

Independent question · withhold the key

Two recordings contain the same word: one quiet and one loud. Which device receives the sound? Does a larger wave automatically mean a different word? Explain.

Expected independent answer

Teacher key · Expected independent answer

A microphone receives the sound. Larger **amplitude** can mean louder sound; it does not by itself establish a different word. Word recognition needs the sound pattern over time.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Louder always means a different word.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

No. Recorded **amplitude** depends on distance, gain and background as well as source sound. Compare sounds at the same distance and gain in similar conditions. Even then **amplitude** does not identify the words or their meaning.

Explanations for the additional workbook examples

W1 Read samples and gaps

Workbook question · W1

Five **sample**s have values 0,3,0,1,0 at times 0,2,4,6,8 milliseconds. State the **sample** count, interval and elapsed time from first to last. Explain why there are four gaps.

Teacher reasoning · W1

Five **sample**s, a 2 ms interval, and 8 ms from first to last. Consecutive pairs make gaps 0-2, 2-4, 4-6 and 6-8: four gaps between five **sample**s.

W2 Inspect what was discarded

Workbook question · W2

Retain only times 0,4,8 from W1. Write the retained values. Give one claim about the original **signal** that these retained values do not justify.

Teacher reasoning · W2

The retained values are 0,0,0. They do not justify saying the **signal** was always zero or had no changes between those times; the original had values 3 and 1 at omitted times.

W3 Compare equal summaries

Workbook question · W3

Sequences A=0,2,0,2,0 and B=0,0,2,0,0 use the same time interval. State their maximum values and one difference the maximum discards. Name information that would preserve that difference.

Teacher reasoning · W3

Both maxima are 2. A reaches 2 at two **sample** positions while B does so at one; their timing patterns differ. The full ordered **sample**s with timing preserve the distinction. A maximum alone does not.

W4 Construct a new hidden change

Workbook question · W4

A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-**sample** sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times.

Explain why the retained record cannot prove that nothing changed between measurements.

Teacher reasoning · W4

One pair is [0,0,0,0,0,0,0] and [0,2,1,0,3,1,0]. Both have zero at indices/times 0,3,6, but the second changes at omitted times.

The retained record is compatible with both, so it cannot establish an unchanged intervening **signal**. Any valid seven-value pair with those constraints is acceptable.

W5 Check a speech pipeline

Workbook question · W5

A recording says Set the timer for fourteen minutes. A **transcript** says forty minutes and the timer is set to 40.

Name the first mismatch, propose a confirmation step and explain why the **transcript** cannot be treated as a complete copy of the recording.

Teacher reasoning · W5

The first stated mismatch is fourteen becoming forty in **transcript**ion. Show or read back the interpreted duration and require correction/confirmation before starting the timer.

A **transcript** may misstate words and omit tone, pauses and background sounds; correct handling of 40 does not validate the earlier **transcript**ion.

Assess independent transfer

Independent question · withhold the key

Selected case: A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-**sample** sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times.

Explain why the retained record cannot prove that nothing changed between measurements.

Selected case answer

Teacher key · Selected case answer

One pair is [0,0,0,0,0,0,0] and [0,2,1,0,3,1,0]. Both have zero at indices/times 0,3,6, but the second changes at omitted times.

The retained record is compatible with both, so it cannot establish an unchanged intervening **signal**. Any valid seven-value pair with those constraints is acceptable.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Sound Becomes a Signal to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

If the sorter uses sound, specify the exact sound pattern needed and test whether classroom **noise** makes that **input** unreliable.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 7: Categories and Labels. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use large labelled boxes and tactile counters. Read values aloud while pointing to their time labels. No hearing-based task is required.

Extension: ask whether doubling the interval changes the number of **sample**s, the first-to-last duration or the stored values when the same list is replayed under the new timing convention.

W4 is secure when each proposed sequence has seven values, both satisfy the zero constraints at 0,3,6, at least one omitted position differs and the explanation identifies ambiguity between measurements.

If positions are confused, reteach with labelled time boxes, then assess a fresh retained-time pattern.

Week 7 builds on this unit by defining categories operationally so different people can label the same **evidence** consistently.

Plan the delayed return

Suggested return points in this level: ai-1/7, ai-1/9, ai-1/12. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Record the sorter's **sampling interval** and retain relevant sequences when a decision depends on changes over time.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Sound Becomes a Signal lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Interpret an ordered sequence of sound-**signal sample**s with a stated time interval

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Retain only times 0,4,8 from W1. Write the retained values. Give one claim about the original **signal** that these retained values do not justify.

Supported practice answer

Reasoned practice answer: The retained values are 0,0,0. They do not justify saying the **signal** was always zero or had no changes between those times; the original had values 3 and 1 at omitted times.

Check every learner before discussion: Use the worked case for Sound as **sample**d information. Proposed explanation: “Five **sample**s make five gaps”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Place five cards in a row and point to each space between them. The span ends at the last **sample**.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Retain only **sample**s at 0,2,4 ms. Require a **prediction** and a reason before comparing. Expected result: Both retained sequences are 0,0,0. Reason: The selected times omit both nonzero **sample**s from each **signal**.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

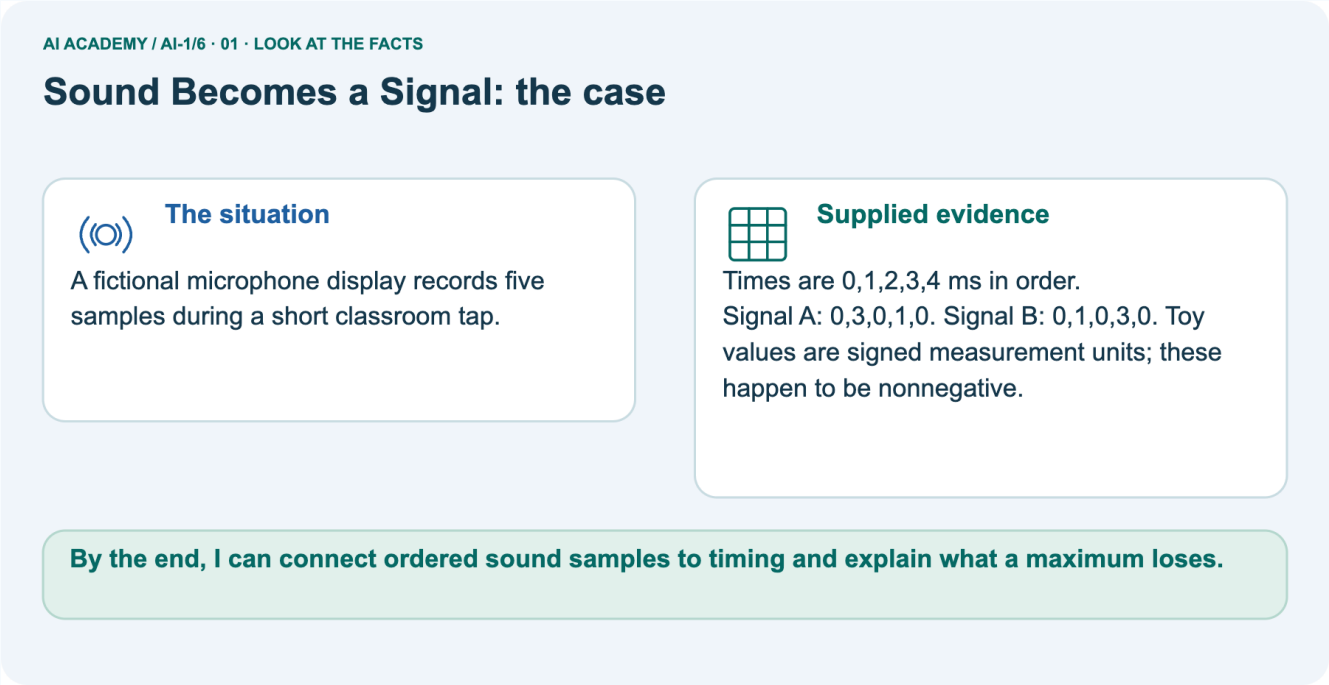
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Sound Becomes a Signal graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Sound Becomes a Signal: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional microphone display records five **sample**s during a short classroom tap. Supplied facts: Times are 0,1,2,3,4 ms in order. **Signal** A: 0,3,0,1,0. **Signal** B: 0,1,0,3,0.

Toy values are signed measurement units; these happen to be nonnegative.

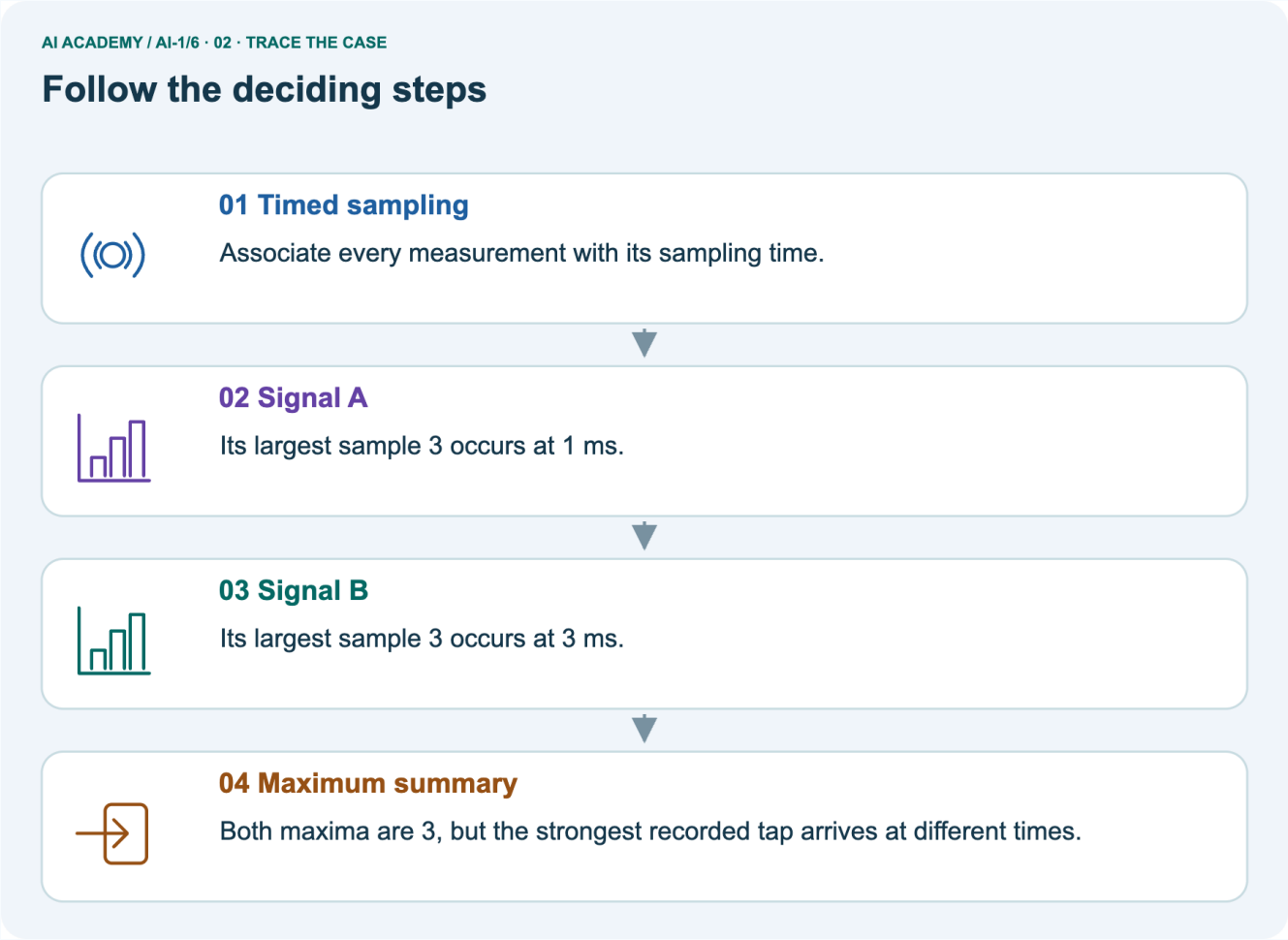
> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent **transcript** if the

image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

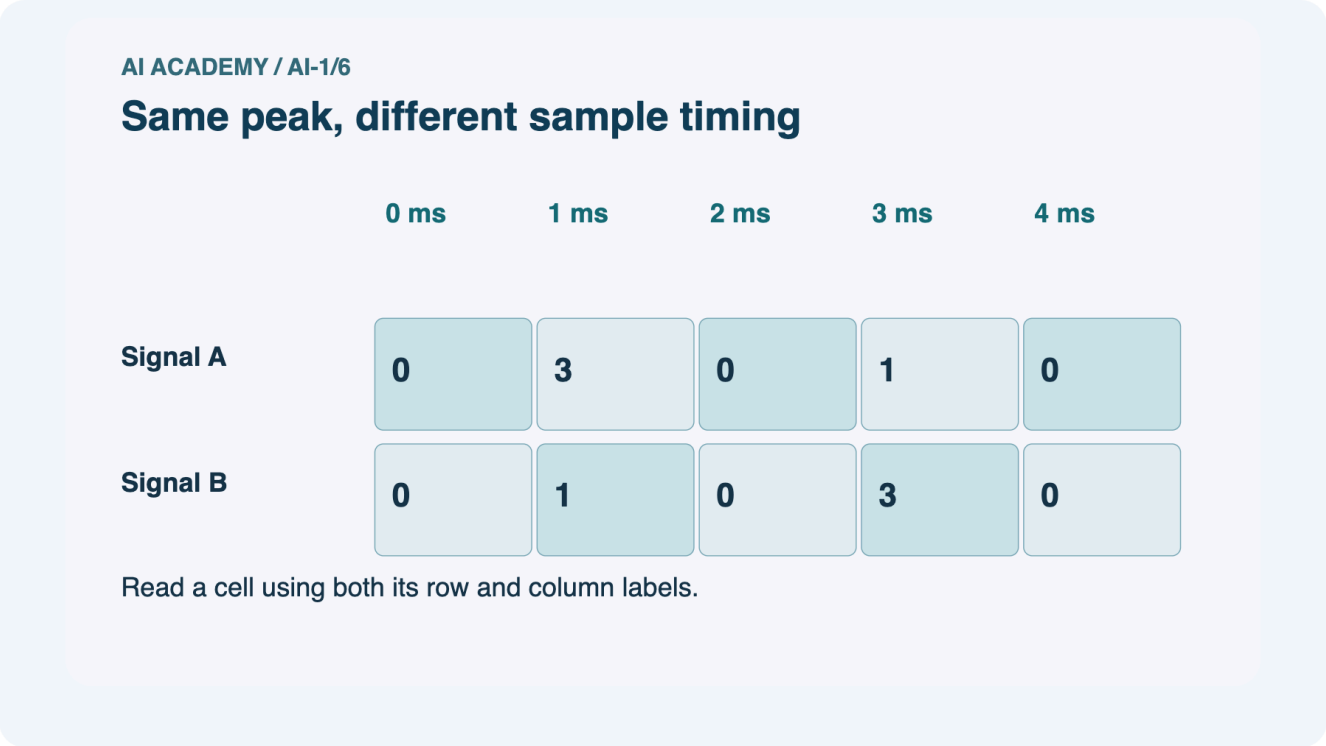
Timed sampling: Associate every measurement with its sampling time. → **Signal** A: Its largest **sample** 3 occurs at 1 ms. → **Signal** B: Its largest **sample** 3 occurs at 3 ms. → Maximum summary: Both maxima are 3, but the strongest recorded tap arrives at different times.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent **transcript** if the image is difficult to see.

03. Same peak, different sample timing



Follow the labelled picture with the matching explanation. Open the vector diagram

Same peak, different **sample** timing. **Signal** A: 0 ms = 0, 1 ms = 3, 2 ms = 0, 3 ms = 1, 4 ms = 0. **Signal** B: 0 ms = 0, 1 ms = 1, 2 ms = 0, 3 ms = 3, 4 ms = 0.

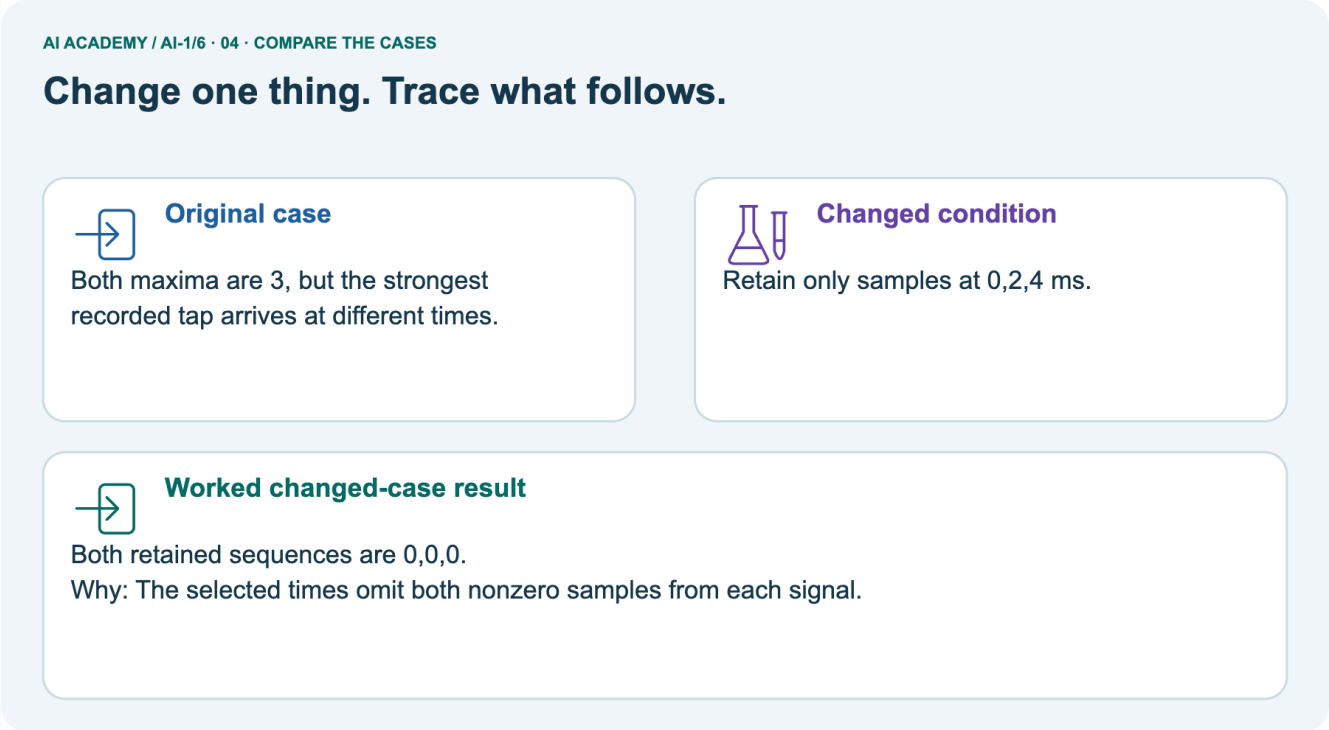
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent **transcript** if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both maxima are 3, but the strongest recorded tap arrives at different times. Changed condition: Retain only **sample**s at 0,2,4 ms. Expected result: Both retained sequences are 0,0,0.

Reason: The selected times omit both nonzero **sample**s from each **signal**.

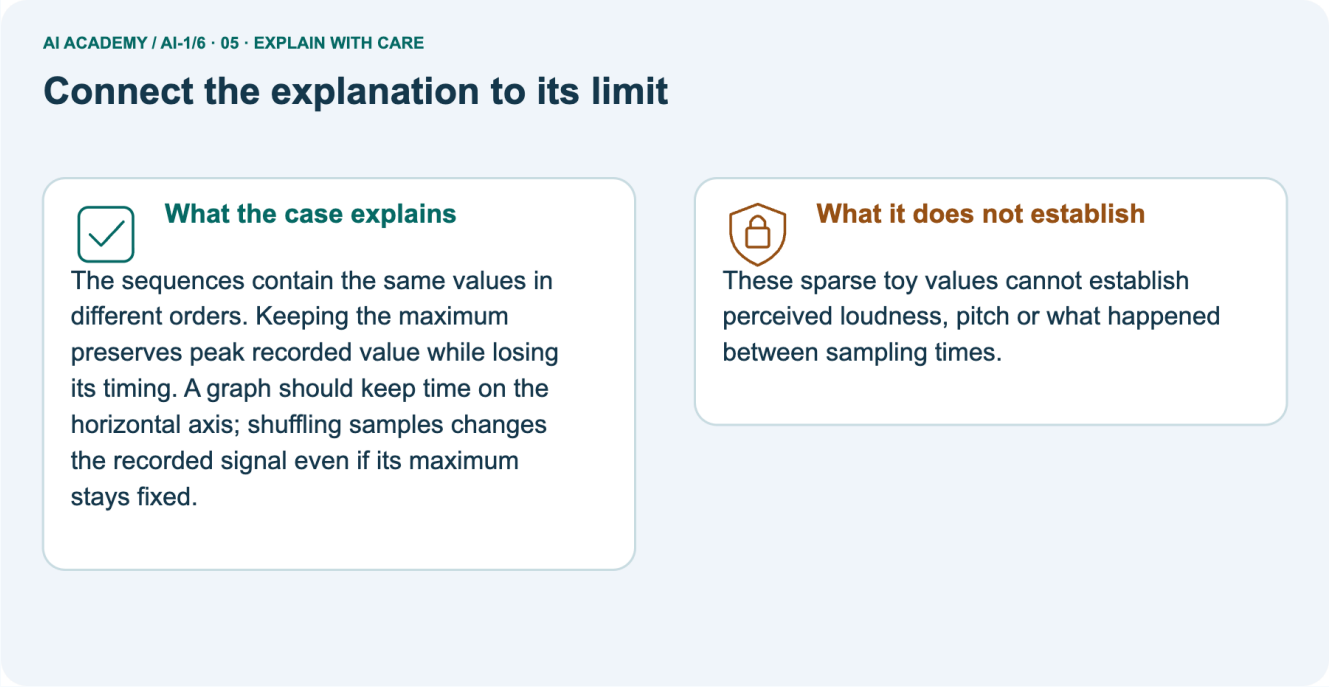
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent **transcript** if the image is difficult to see. Changed-case reasoning: The selected times omit both nonzero **sample**s from each **signal**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The sequences contain the same values in different orders. Keeping the maximum preserves peak recorded value while losing its timing.

A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

Limit: These sparse toy values cannot establish perceived loudness, pitch or what happened between sampling times.

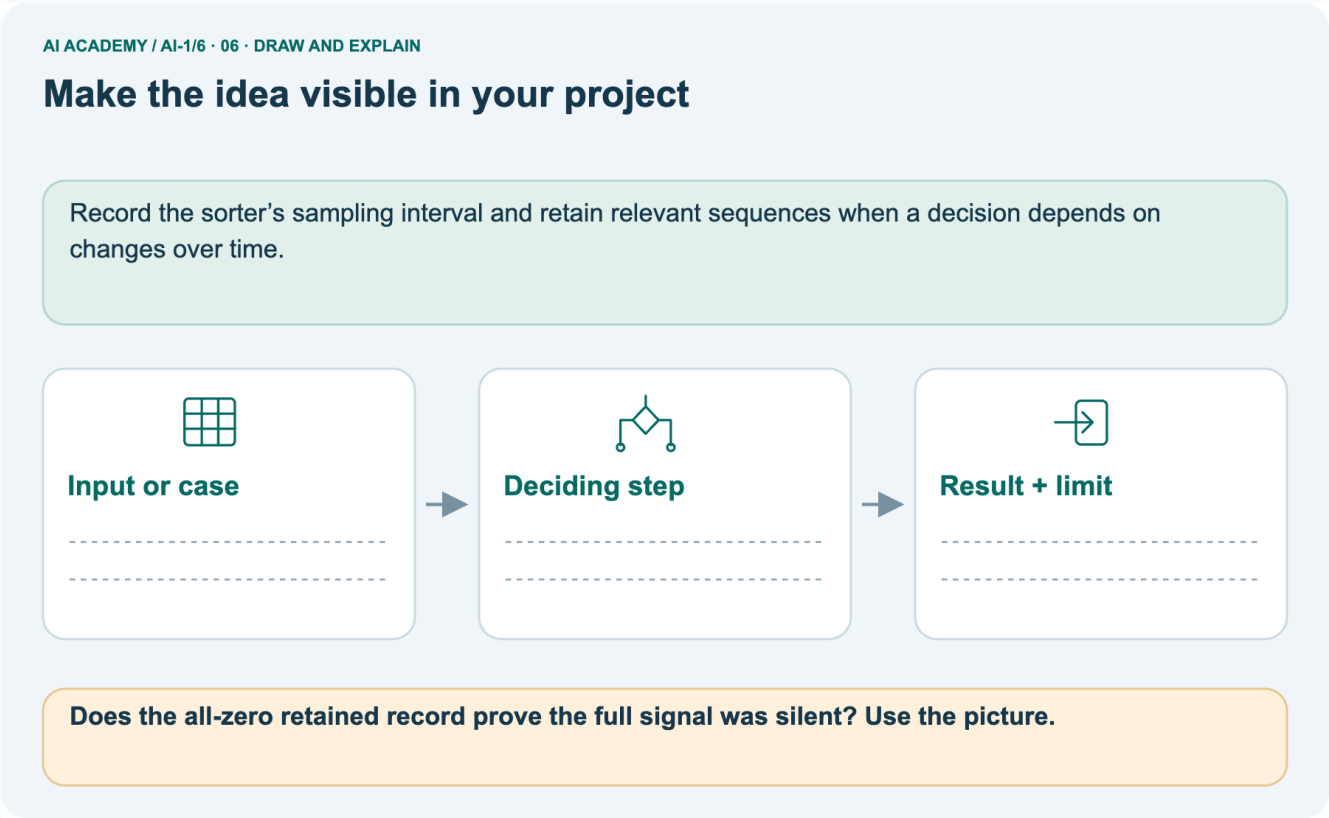
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent **transcript** if the image is difficult to see.

Keep the stated scope: These sparse toy values cannot establish perceived loudness, pitch or what happened between sampling times.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Record the sorter's **sampling interval** and retain relevant sequences when a decision depends on changes over time. Draw your **input**, deciding step, result and limit.

Supported question: Does the all-zero retained record prove the full **signal** was silent? Use the picture.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent **transcript** if the image is difficult to see.

> Supported practice key: No. The full supplied **signal**s contain nonzero values at 1 and 3 ms; those measurements were discarded.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.